4.2.1 Current and conventional current direction

Current is charge passing a cross-section per unit time; 1 A is 1 C per second. Conventional current follows positive-charge motion, from the positive to negative terminal through the external circuit. Electrons in metals move oppositely. Components convert electrical energy, but do not use up the charge flowing through them.

$$I_{\text{av}} = \frac{Q}{\Delta t}$$

I_{av}: average current (A); Q: charge passing a cross-section (C); Δt: time (s). Equals current for steady flow.

4.2.2 Series current and junction conservation

In a steady circuit, charge does not continually accumulate. Current is therefore the same everywhere along one series branch, and total current entering a junction equals total current leaving it. Branch currents need not be equal. They depend on branch resistances and voltages. Energy dissipation in a resistor does not consume current.

4.2.3 Electromotive force

A source's electromotive force, or e.m.f., is energy supplied per unit charge and is measured in volts. It is not a force. Internal resistance dissipates some energy when a real source supplies current, so terminal voltage can be lower than e.m.f. With no current, terminal voltage is approximately equal to e.m.f.

$$\varepsilon = \frac{E_{\text{supplied}}}{Q}$$

ε: source e.m.f. (V); E_{supplied}: energy supplied by the source (J); Q: charge moved (C). Energy per charge, not a force.

4.2.4 Potential difference

Potential difference relates energy change to the charge transferred between two points. For a load, its magnitude can be found by dividing electrical energy converted by charge passing through. One volt is one joule per coulomb. Voltage is a comparison between points, not something that flows through a component.

$$V = \frac{E_{\text{transferred}}}{Q}$$

V: potential difference (V); $E_{\text{transferred}}$: electrical energy converted in the load (J); Q: charge (C). For a signed potential change use ΔV = ΔU/q.

4.2.5 Voltages in series and parallel

Series loads share the applied terminal voltage, with individual voltage drops adding to the total. Parallel branches connected to the same two nodes have equal potential differences. Series voltage division is not necessarily equal and depends on resistance. With source internal resistance, the external terminal voltage must not automatically be equated to e.m.f.

4.2.6 Cells in series and parallel

Series-cell e.m.f.s add according to polarity: aligned cells add and opposing cells subtract. Identical cells connected in parallel with matching polarity have the same total e.m.f. as one cell, but a smaller equivalent internal resistance. Cells with unequal e.m.f.s can produce circulating currents and must be analysed separately.

4.2.7 Resistance and Ohm's law

Resistance at an operating point is the voltage-to-current ratio, $R = V/I$. An ohmic component has $V \propto I$ and constant resistance when temperature and other physical conditions remain fixed. A filament's resistance rises as it heats, so its graph can curve.

With V on the vertical axis and I on the horizontal axis, the slope of a straight line through the origin equals resistance. For a curved graph, read V and I at the required point and calculate V/I. This is the slope of the line joining the origin to that point. Reversing the axes makes the straight-line slope the reciprocal of resistance.

$$R = \frac{V}{I}$$

$$V \propto I$$

R: resistance (Ω); V: voltage across component (V); I: current through it (A). Definition at an operating point; constant R additionally requires ohmic behaviour and fixed physical conditions.

Ohm's law: temperature and other physical conditions remain constant. V: voltage; I: current.

4.2.8 Factors affecting wire resistance

For a uniform wire, resistance is proportional to length and resistivity and inversely proportional to cross-sectional area. Longer wires resist more. Thicker ones resist less under comparable conditions. Resistivity depends on material and temperature. Convert radius or diameter correctly before calculating area. Stretching at constant volume changes both length and area.

$$R = \frac{\rho l}{A}$$

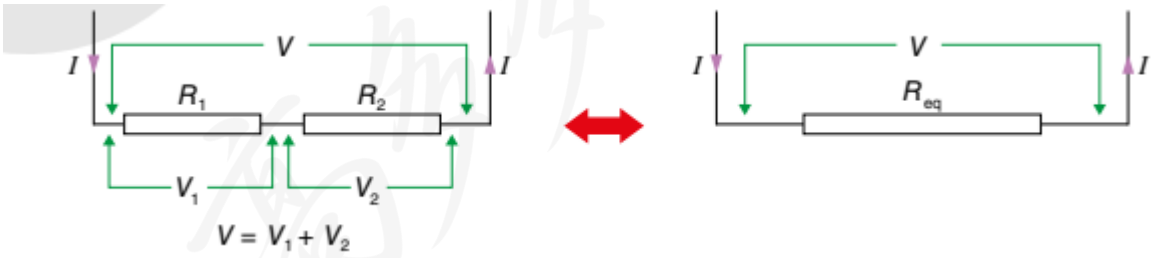
Uniform wire: R (Ω); resistivity ρ (Ω m); length l (m); cross-sectional area A (m²). Temperature fixed.

4.2.9 Resistors in series

Series resistors carry the same current and their voltage drops add, so equivalent resistance is the sum of individual resistances. Adding a series resistor increases total resistance and reduces current at fixed terminal voltage. With the same current, the larger resistor has the larger voltage drop. Voltage is not necessarily shared equally.

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$$

Resistors in series carry the same current. R_{eq}: equivalent resistance (Ω); R_{i}: individual resistances (Ω).



The equivalent circuit for series resistors: the same current passes through the original branch.

4.2.10 Resistors in parallel

Parallel resistors have equal potential differences and their branch currents add, giving the reciprocal-sum rule for equivalent resistance. The equivalent is smaller than every individual finite positive branch resistance. For exactly two branches, the product-over-sum shortcut applies. It cannot be used unchanged for three or more resistors.

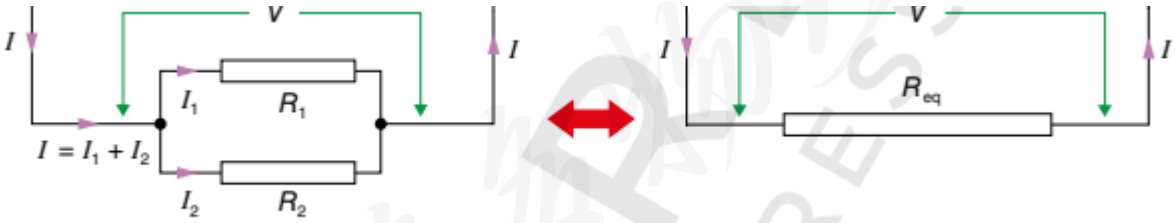
$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

Exactly two resistors in parallel, with the same voltage across both; all resistances in Ω.

Parallel branches share the same voltage. R_{eq} and R_{i} in Ω; reciprocal resistances add.



Parallel branches share the same two nodes and therefore have equal potential differences.

4.2.11 Power: the rate of energy transfer

Power measures energy transferred or converted per unit time. Average power is the total energy transferred divided by the elapsed time; 1 W means 1 J per second. Transferring the same energy in less time requires greater average power. If power varies, this average need not equal the instantaneous power at any chosen moment.

$$P_{\text{av}} = \frac{E}{\Delta t}$$

P_{av}: average power (W); E: energy transferred (J); Δt: time (s).

4.2.12 Electrical power and resistive heating

Electrical power is voltage times current. For a resistive load, substituting the resistance relation gives current squared times resistance or voltage squared divided by resistance. Decide whether voltage or current is fixed before comparing powers: increasing resistance has opposite effects in these cases. Alternating-current mean power requires the applicable r.m.s. relations.

$$P = VI$$

$$P = VI = I^2R = \frac{V^2}{R}$$

P: electrical power (W); V: p.d. (V); I: current (A). Steady DC values or simultaneous instantaneous values.

Resistive load; P: W; V: V; I: A; R: Ω. Use DC values, or RMS voltage/current for a constant resistor in AC.